مدارس الإمارات الوطنية
Emirates National Schools

Grade: 5 | Term 1, Week 13 (23 Nov to 27 Nov) | TCD: Future Skills 5: Prototyping and Smart Making | Session length: 1 hr 30 min

T1 WEEK: 13	**Lesson 1: Finalise and 3D Print**
Standard(s):	Technology: CAD to 3D print workflow; Math: 5.MD.C (volume), measurement and precision
Literacy Strategies:	Print-ready checklist with picture prompts; vocabulary cards for STL, slicer, layer, infill and support.
Assessment:	Print-ready checklist; exported STL; print evaluation; live quiz on the TCD website (counts towards the Junior Designer Challenge certificate).
WALT (Learning Target):	We are learning to prepare a 3D model so it prints well, and to check the print against our plan.
WILF (Success Criteria):	I can check my model with the print-ready checklist. I can export my model as an STL file. I can explain what the slicer does, including layers, infill and supports. I can compare my print with my design and suggest one improvement.
Targeted Vocabulary	STL, export, slicer, layer, infill, support, nozzle, filament, print time
Resources	Computers or iPads with Tinkercad, 3D printer and PLA filament, Slicer software on the teacher's computer, Print-ready checklist, Calipers or rulers, Sample prints showing good and poor quality Accompanying resource: Grade5_Week13_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade5_Week13_Worksheet and Grade5_Week13_Lesson_Plan.
National Identity Integration	Link to Dubai's 3D Printing Strategy and the 3D printed Office of the Future in Dubai, showing how the UAE uses 3D printing in real buildings.
AI/Digital Learning	Tinkercad; slicer software (for example Ultimaker Cura) on the teacher's computer; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Go through the print-ready checklist using a model with problems (slide 6). Model exporting as STL (slide 7). Demonstrate the slicer on the projector (slide 8). Go through printer safety (slide 11).
Guided Practice: We Do	Partners check each other's model with the checklist and point out one fix.
Independent Practice: You Do	Students finalise and export their model (slides 12 and 10). Teacher slices and starts prints in batches; students evaluate prints as they finish (slides 9 and 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Use the checklist with a partner and export with the step card.** **On: Fix all checklist problems and explain layers and infill.** **Above: Compare two infill settings in the slicer demo and explain which you would choose and why.**
Closure/Reflection	Prints that are ready are shown. Students compare with the reference. Exit ticket (slide 15). Prints not finished will be collected next week.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☐ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

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G&T / Extension	☐ Extension / Challenge Task ☑ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☑ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____